

Starting and Shipping Your Project

Autumn term 2026

Practical advice, not rules. The binding rules are in the Requirements document; nothing here changes them. If the two ever disagree, the Requirements win.

Two halves. **Part I** is about starting: choosing a question, finding data, sizing the project, writing the proposal. **Part II** is about shipping — handing over work that someone else can open, read and run without asking you a single question. Most marks lost on this project are lost in Part II, not in the modelling; most projects that never get going were lost in Part I.

Part I — Starting

What a project is

One question, one data set you can actually get, one or more methods from the course applied to it with judgement, and a paper that lets a stranger reproduce what you did. That is the whole thing. The grade rewards evidence that you can apply what the course taught — not the size of the data, not the number of models, and not a positive result.

The shape that works, in one line: *"Does X predict / explain / separate Y in data Z?"* If you can fill in X, Y and Z today, you have a project. If you can only fill in one of them, you have a topic, and the next section is for you.

Choosing a question

Three tests, in order. A question that passes all three is ready to propose.

1. **Can you say it in one sentence?** "I want to look at cryptocurrencies" is a topic. "Do Bitcoin returns lead Ethereum returns at the daily horizon?" is a question — it names the data, the relationship, and what an answer would look like.
2. **Can you get the data this month?** Not "there must be data on this", but a file on your disk, or a download script that runs. Most proposals that fail, fail here. Find the data *before* you fall in love with the question.
3. **Does a method from the course fit it?** Regression and classification are taught by mid-October and cover most questions; neural networks and recurrent models follow in early November; Gaussian processes, dimensionality reduction and clustering land in the second half of November. Plan around when the method you need arrives, and remember that the simplest method that fits is the right first model.

Three shapes of project, each of which has produced good work:

- **Predict a quantity.** Rents from municipality characteristics; quarterly GDP from monthly indicators; option prices from their inputs; tomorrow's volatility from this week's. Regression, then something non-linear if the linear model leaves structure behind.
- **Classify an outcome.** Loan default from applicant data; whether a central-bank an-

nouncement day moves a stock more than a normal day; recession or not from the yield curve. Logistic regression, trees, k-nearest neighbours, a small network.

- **Find structure nobody labelled.** Cluster countries by their macroeconomic path since 2000; reduce a hundred indicators to the three that carry the variance; find regimes in a price series. Principal components, k-means, mixture models.

These are illustrations, not a menu. The best proposals we receive are questions the student was already curious about, brought to data.

What makes it yours. A question is original when it is your own, or when a known question is asked of data nobody asked it of — a different country, market, period or sample. It does not need to be novel to the literature. It does need to be more than an exercise from the course with the numbers changed, and more than a tutorial re-run on a dataset that came clean.

Finding data

Places econ and finance students find usable data, all free:

- **Macro and official statistics:** FRED (fred.stlouisfed.org), the World Bank (data.worldbank.org), OECD (data-explorer.oecd.org), Eurostat (ec.europa.eu/eurostat), the ECB Data Portal (data.ecb.europa.eu), the SNB data portal (data.snb.ch), the Swiss Federal Statistical Office (bfs.admin.ch), Our World in Data (ourworldindata.org).
- **Markets:** daily prices through the `yfinance` package; company filings on SEC EDGAR (sec.gov/edgar); Kaggle (kaggle.com/datasets) for curated sets of almost everything, with the caveat below.
- **Reference collections:** the UCI Machine Learning Repository (archive.ics.uci.edu) and the datasets built into scikit-learn — fine for a first model, weak as the whole project, because they arrive clean and everyone has seen them.

Four rules, learned the hard way:

- **Download it now and keep a copy.** Websites change, APIs get rate limits, free keys expire. A CSV in your repository on 20 December is worth more than an API that worked in October.
- **Check the licence before you commit to it.** You will be publishing the repository. Public-domain and open data are fine; scraped or licensed data needs the download script instead of the file (Requirement 4.15), and you must be allowed to use it at all.
- **Messy is good.** Missing values, inconsistent units, a merge across two sources, a variable that needs constructing — this is where the data-preparation work lives, and that work is graded. A dataset that needs no cleaning has taken part of the project away from you.
- **Small is fine.** A few thousand rows is plenty for every method in this course. Choose the data by the question, not by its size.

How big a project is

This is a course in machine learning, and the project is where you show what you took from it. There are two good ways to do that.

One method, taken all the way. Fitting a model and reporting its score is where this kind of project starts, not where it ends. Taking a method all the way means checking its assumptions against your data, tuning it and saying why, looking closely at where it gets things wrong and what that tells you about your question, and testing whether the result survives a reasonable change to the setup.

Several methods, each with a job. A simple model first, then a second one brought in

because the first showed you something it could not handle, and a comparison that answers a question rather than a table of scores. Where your question allows it, draw on the course beyond its first weeks: regression is where the course starts, not where it ends.

Either way, what we look for is the same: decisions that are yours, and that you can explain. Three models fitted carelessly show less than one fitted with care. The grade follows what the project shows, not the hours that went into it.

Two ways projects come up short. The first is the tutorial re-run — a clean dataset, a default model, a plot, and no decision anywhere that was the student's own. The second is the project that outgrows its owner: too much data, a method not yet understood, or compute the workspace does not have. The proposal exists to catch both while they are still cheap to fix.

Everything must run in your Nuvolos workspace (Requirement 4.22). Part II says what that means in practice; the short version is that every method this course teaches fits comfortably, and if your idea needs a GPU or a two-day parameter search, it is the wrong idea.

The proposal

Two hundred words, three things (Requirement 3.1): the question, the data, the methods. A template that fits:

Question. One or two sentences stating what you want to find out and why it is interesting. **Data.** What it is, where it comes from, how much of it there is, and — the sentence that gets proposals approved — that you already have it. **Method.** Which methods from the course you will use, in what order, and what result would count as an answer either way.

An example, to show the register rather than the topic:

Does the yield-curve slope predict Swiss recessions at a one-year horizon, and does a non-linear model add anything over the standard probit? I have quarterly Swiss GDP and SNB term-structure data for 1990–2024 (SNB data portal and FSO, 140 quarters, downloaded). I will define recessions by two consecutive negative quarters, fit a logistic regression on the 10-year–1-year spread and two controls, then compare it with a k-nearest-neighbours classifier and a small neural network on the same features under time-ordered cross-validation. The result is a comparison of out-of-sample classification performance; if the simple model wins, that is the finding.

Proposals get sent back for one of four reasons: the data is not in hand; the question cannot be answered with the data named; the scope will not fit the workspace or the calendar; or it is an exercise from the course with the labels changed. Each is a five-minute fix at proposal stage and a lost fortnight in December.

Send it early. The deadline, 9 November, is the last day a proposal is accepted — it is not the day to aim for. Approval is by return email and is given as proposals arrive, so a proposal sent in mid-October is approved in mid-October and buys you three extra weeks of work. A proposal sent on 9 November leaves six weeks for everything.

A timeline that fits the course

- **Weeks 1–3 (14 Sep – 4 Oct)** — *Introduction; Python, notebooks, GitHub.* Set up the accounts below. Pick a domain you are curious about. Find and download candidate data. Look at it.
- **Weeks 4–6 (5 – 25 Oct)** — *Regression, classification.* Fit the first, simplest model to your data. Write the proposal. **Send it.**

- **Weeks 7–8 (26 Oct – 8 Nov)** — *Neural networks, RNNs*. Proposal approved; the data pipeline is a notebook that runs top to bottom. Add the second method if you planned one.
- **Weeks 9–12 (9 – 30 Nov)** — *Recap; Gaussian processes; PCA; clustering*. The main work. Write the paper as you go — a section a week, not a paper in a week.
- **Week 13 (7 Dec)** — *Reinforcement learning*. Results final. Figures made by code. Record the presentation.
- **Week 14 (14 – 20 Dec)** — *Wrap-up*. Reproduction test (Part II). Submit by **Sunday 20 December, 23:59**.

This week

Before the first lecture, or in its first hour. None of this needs the course to have started:

1. Read the Requirements document once, end to end. It is short and it is binding.
2. Put your full name on your GitHub account, and create an empty repository for the project (see Part II, first section, for why both matter).
3. Open the SIAM template in Overleaf and save it as a project. You will write into it from October.
4. Enrol in the Nuvolos workspace with the email you use for Moodle, then log in once so you know the login page from the enrolment page.
5. Start a note titled "data I could use". Every dataset you come across in the next month goes in it, with its URL. By week 4 the note will contain your project.

Part II — Shipping

Two things to do in the first week

Put your real name on your GitHub account. Requirement 4.13 asks for your full name in the account's profile name, and Requirement 4.14 says a submission that cannot be matched to a registered student is not graded. Most student accounts are called something else. Changing it takes ten seconds in Settings → Public profile, and it is the single cheapest way to avoid losing a grade you have earned.

Make the repository now, not in December. An empty repository with a README costs nothing and means the last week is about the work.

The test that matters

Before you submit, do this on a machine that is not the one you worked on:

1. `git clone` your own repository into an empty folder.
2. Follow your own README exactly as written, changing nothing.
3. See whether the results in your paper appear.

If step 3 fails, you have found what the marker would have found. Almost every problem below is caught by this one test. Cloning is the honest version of the test: it shows you only what you actually committed, not what happens to be sitting on your desk.

What the repository holds

paper.pdf
README.md

the research paper
how to reproduce, in order

<code>requirements.txt</code>	the packages needed
<code>recording.mp4</code>	or a link in the README, if the file is large
<code>code/</code>	
<code>01_get_data.py</code>	or <code>.ipynb</code> - one stage each
<code>02_clean.ipynb</code>	
<code>03_model.ipynb</code>	
<code>04_figures.ipynb</code>	
<code>helpers.py</code>	anything used more than once
<code>data/</code>	the data, or the script that fetches it
<code>figures/</code>	what the paper shows

Numbering the files is not required, but it answers "what do I run first?" without anyone having to ask.

Use lowercase names with no spaces and no accents. `data final (2).csv` breaks on other people's machines; `prices_2020_2024.csv` does not.

One notebook per stage

Requirement 4.8 says no single file may contain the whole analysis. This is the requirement people most often ignore, so here is why it exists.

A 5000-line notebook that loads, cleans, models and plots cannot be checked. Nobody can run part of it, nobody can find the line that made a number, and nobody — including you, in January — can tell which cells were still live and which were abandoned three weeks ago. It also makes Requirement 6.1 impossible to satisfy: in one enormous file, no reader can tell which parts you wrote.

Split by stage instead: get the data, clean it, model it, make the figures. Four files. Each one starts by loading what the previous stage saved and ends by saving what the next stage needs.

Rules of thumb, not requirements:

- If a notebook takes more than a couple of minutes to scroll through, it is doing more than one job.
- If you have copied a function into a second notebook, it belongs in `helpers.py` (Requirement 4.9).
- If a cell only works when you run the cells above it in a particular order that is not top to bottom, the notebook is broken. Restart the kernel and run all, before you commit.

Keep the repository clean

Requirement 4.12 excludes caches, checkpoints, virtual environments and editor settings. A `.gitignore` handles all of it:

```
__pycache__/  
.ipynb_checkpoints/  
.venv/  
.DS_Store  
.vscode/
```

Add it in the first week and you never think about it again.

Working with the repository

- **Commit as you go.** A commit a day is a record of your work. One enormous commit on the last evening is a record of nothing, and it is also when things go wrong.
- **What counts is the last commit before the deadline** (Requirement 4.12). Committing is not the same as pushing: run `git push` and then look at the repository page in your browser to confirm your work is actually there.
- **Do not commit very large files.** GitHub rejects anything over 100 MB and gets slow well before that. If your data is large, Requirement 4.15 lets you commit the download script instead.
- **Use a `.gitignore`** for caches, checkpoints and virtual environments (`__pycache__/, .ipynb_checkpoints/, .venv/`). They are noise, and they make the repository heavy.
- **Never commit a key or a password.** Deleting it in a later commit does not remove it: it stays in the history. If it happens, tell us and revoke the key.

The README

Half a page is plenty. In this order:

- what the project does, in two sentences;
- what to install (`pip install -r requirements.txt` is the whole answer, if you write the requirements file);
- what to run, in the order to run it;
- how long it takes, and what it produces;
- where the data came from.

Write it as instructions to a stranger, because that is who reads it.

Code

- **A script that runs top to bottom beats clever code that needs explaining.** You are graded on evidence that you can apply the methods, not on style.
- **Set the random seed.** If your numbers change every run, nobody can check anything in your paper.
- **No absolute paths.** `/Users/anna/Desktop/thesis/data.csv` exists on exactly one computer in the world. Use `data/prices.csv`.
- **Restart the notebook and run it once, in order, before submitting.** Notebooks that work only if cells are run out of order are the most common reproducibility failure there is.
- **Keep secrets out.** No API keys in the code. If your project needs one, say in the README which key is needed and where to put it.

Data

- Small and shareable: include it.
- Large or licensed: include the download script and say plainly what the licence permits. Requirement 4.15 allows this.
- Say where every data set came from and what you did to clean it. "I dropped 412 rows with missing prices" is a sentence the marker wants to read; silence about it is not.

The paper

- **Open the template on day one**, not in December: Template for SIAM Journals → **Open as Template**. It creates an Overleaf project in your account. Nothing to download, nothing to install, and it compiles in the browser. Writing into the template from the first paragraph costs nothing; reformatting a finished paper into it costs an evening.
- If you have never used LaTeX: you do not need to learn it properly. Replace the title, the author, and the section bodies. The template already handles the rest.
- Overleaf's free tier is enough for this project. If you would rather write elsewhere, that is allowed — Requirement 4.4 asks for the output format, not the tool.
- Every figure needs axis labels, units, and a caption saying what to look at.
- Report what you actually ran. If a method failed, that belongs in the paper — under Requirement 7.3 an honest negative result is not held against you.

The recording

- The recording is the presentation — there is no live one, and no day to be absent on. It is due with everything else.
- Fifteen minutes, screen plus voice. A Zoom recording of yourself is fine.
- A structure that fits the time: the question (2 min), the data (2 min), the method (4 min), the results (5 min), what you would do next (2 min).
- Watch the first minute back before submitting. Inaudible sound is the usual failure, and it cannot be fixed after the deadline.
- Say your name at the start.

Compute: what you get, and scoping to it

You get a Nuvolos workspace, free, for the whole course. That is the only compute the University provides for this project — there are no cloud credits, no GPU allocation, and no licences beyond what is already in the workspace.

The workspace is modest and deliberately so: it is enough for the methods this course teaches, applied to data of a sensible size. Almost every good project fits in it comfortably.

Scope to it at proposal stage, not in December. If your idea needs to fine-tune a large model or grid-search for two days, it is out of scope (Requirement 4.22), and the proposal is where we say so — which is the whole point of asking for a proposal. Bring the question then, not after you have spent three weeks on it.

Ways a project shrinks to fit without becoming trivial:

- take a sample of the data rather than all of it, and say so in the paper;
- use fewer, better-chosen parameter settings instead of a large search;
- pick the smaller model. A well-understood small model beats a large one you could not inspect.

If you use your own compute, you may — at your own cost, and it earns you nothing. Requirement 7.2 assesses the work, not the hardware. Two practical obligations come with it: say in the README what you ran it on and roughly how long it took (Requirement 4.24), and if the result cannot be reproduced in the workspace, commit the code, the saved output, and a note of what produced it (Requirement 4.25). A number nobody can trace back to code is not evidence.

And note Requirement 4.26: a machine you rented failing, or running out of credit, is not a

reason for an extension. The workspace you were given does not have that failure mode.

Code you did not write

Section 6 of the Requirements is the one people get wrong, so here is the short version.

You do not attribute libraries. `import pandas` is using a tool.

You do attribute anything you did not write yourself: a function from Stack Overflow, a plotting recipe from a blog, a chunk of a repository, a replication package from a paper. One comment where it appears is enough:

```
# adapted from https://github.com/someone/repo, MIT licence
def rolling_sharpe(returns, window):
```

```
...
```

Then list it in the appendix with the licence. That is the whole job, and it takes about a minute.

Where it gets awkward: assistants. When ChatGPT writes a function, it sometimes reproduces code that already exists somewhere, and does not tell you. You cannot check this, and you are not expected to. What you can do is say the function came from an assistant and what you asked for. If someone later recognises the code, that declaration is the difference between an honest mistake and a misconduct case. It is a minute's work buying a great deal of protection.

The test to apply: for every block of code, can you say where it came from? If the answer is "no idea, it has been in the folder for weeks", that block cannot be attributed — and Requirement 6.7 says it should not be submitted. Rewrite it or find the source again.

Attribution never costs you marks. Requirement 6.9 is explicit: what is assessed is what you built with the code, and the offence is the silence, not the borrowing.

Declaring your tools

Requirement 5 asks which tools you used and what for. A few honest lines:

ChatGPT: drafted the plotting code in `figures.py`; rewrote three paragraphs of the introduction for clarity. All statistical results are my own work and were checked by hand.

Nobody is penalised for using assistants — the course teaches you to. The penalty is for not declaring them, and for being unable to explain what you handed in.

Submitting: the commit identifier

Requirement 4.17 asks you to email the repository URL **and** the commit identifier — the SHA. That number is what makes your submission fixed: it names one exact state of your work, so nothing you do afterwards can change what is graded, and nothing anyone else does can either.

To get it, once everything is committed and pushed:

```
git log -1 --format=%H
```

That prints 40 characters. Copy the whole thing into the email.

Check it is really on GitHub before you send: open your repository in a browser and find that commit. `git push` failing quietly, or a commit sitting on a branch that is not the default one, is the usual way this goes wrong — and Requirement 4.19 treats a commit that cannot be

found as a submission not made.

You can keep working after you send the email. It changes nothing: the commit you named is the one that is graded. What you must not do is delete the repository, make it private to us, or rewrite the history — that removes the commit you submitted.

The week before the deadline

- Run the reproduction test above. Now, not on the last evening.
- Check the page count against Requirement 4.3.
- Check the eight required sections are present and in order.
- Send the submission email a day early with the SHA of what you have. If you improve the work afterwards, send a second email with a new SHA before the deadline; the last one before the deadline is the one that counts. An email you have already sent is worth more than a perfect one you were about to send.